The output of Panel C is a subgroup-specific *growth regime* H_S, where H_S is a cumulative distribution function (CDF) on $[0, 1]$ for a latent conditional-percentile random variable P_S (SGPc on the 0–1 scale):

$$H_S(p) = \Pr(P_S \leq p), \qquad 0 \leq p \leq 1.$$

In words, H_S describes how subgroup S allocates conditional ranks *within* the baseline dependence template $F_0(\cdot \mid u)$.

Panel D shows the *distribution of* P_S (typically via its density $f_S(p) = H_S'(p)$ when H_S is smooth), not the CDF itself. Thus the "Best-fit" curve is the density corresponding to the fitted CDF H_S (i.e., $P_S \sim H_S$). The dashed grey "Uniform" reference corresponds to the uniform regime $H(p) = p$ (equivalently $P \sim \mathrm{Unif}(0, 1)$), which places no tilt on conditional percentiles relative to the baseline kernel.

Because we restrict H to the Beta family in Panel C, $P_S \sim \mathrm{Beta}(\kappa m, \kappa(1 - m))$: the mean m sets the regime center (reported on the SGPc scale as $100m$) and κ controls concentration (dispersion of latent SGPc).

The vertical reference lines mark the mean SGPc under the uniform regime (50), the generating "True" regime (), and the recovered best-fit regime (). In applications, the primary outputs are H_S (and uncertainty bands) and summary functionals such as mean/median SGPc and bucket membership probabilities.